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Published in:
Event Proceedings

Publication date:
2024

Document Version
Early version, also known as pre-print

Link to publication from Aalborg University

Citation for published version (APA):
Dorland, J., & Jørgensen, M. S. (2024). From Assessment to Innovation: The Role of LCA and Co-Design in Sustainable Development. In Event Proceedings: ISPIM Connects Osaka – Connecting and Empowering Society The International Society for Professional Innovation Management (ISPIM).

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This paper was presented at ISPIM Connects Osaka – Connecting and Empowering Society, Osaka, Japan on 2-4 December 2024.
Event Proceedings: ISBN 978-952-65069-7-5

From Assessment to Innovation: The Role of LCA and Co-Design in Sustainable Development

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Abstract: Businesses increasingly face demands for sustainability and regulatory requirements like the CSRD, prompting environmental assessments of their products and business models. Some organizations find that Life Cycle Assessments (LCAs) as tool alone are insufficient, and the belief that they could drive sustainable innovation on their own is exaggerated. This paper, based on five case studies from diverse sectors, examines how LCA can drive sustainable development when integrated with Life Cycle Management (LCM) and Scandinavian participatory design (PD) methodologies. Based on co-design activities involving stakeholders such as users, managers, and suppliers, we develop a typology business can use as an analytic and strategic guide to leverage LCA and LCM into actionable strategies for sustainable business development. Our findings reveal that LCA, when combined with LCM and PD, not only facilitates environmental improvements but also drives business model innovations, with an iterative co-design process often proving as critical as the LCA outcomes.
Keywords: Life Cycle Assessment (LCA); circular economy; design-driven integration; stakeholder engagement; sustainable development; staging; functional unit; environmental impact; participatory design; co-design.

1 Introduction

Over recent years, we have experienced an increase in interest from SMEs and other organizations seeking assistance to enhance their sustainability. A number of these requests involve conducting environmental assessments like life-cycle assessments (LCA). Often, upon receiving or during the process of conducting these assessments, many organizations struggle to utilize the outcomes effectively, finding them different from their expectations in both conclusion and form or lacking the competences to integrate and use the outcomes. This observation prompted us to make a deeper examination into how organizations could more effectively leverage these types of analyses, leading to the gradual development of a novel framework. This framework integrates environmental assessment with other scientific analytic approaches through a

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design methodology based on Scandinavian participatory design (PD), facilitating the necessary stakeholder engagement during both the analytical and implementation phases, to not only comply with reporting requirements but also using the environmental assessments to drive innovation.

The art of using LCA to make informed business decisions is often called life cycle management (LCM) (Hauschild et al., 2018; Sanchez et al., 2004). We are specifically interested in the integration of different scientific and analytical perspectives in our framework through PD, and how this can support the strategic and operational focus of LCA of how to go from LCA to LCM.

The framework detailed in this paper represents the culmination of over two decades of development and experimentation in integrating engineering design, sociotechnical analyses, and design methods. This evolution began with our work in the Design and Innovation program at the Technical University of Denmark at the turn of the millennium and continued until 2012. Subsequently, we transitioned to Aalborg University, where we initiated the program in Sustainable Design. In recent years, our focus has expanded from screening-based environmental assessments to including full LCAs, enriching our comprehensive approach. Further details on integration with design are provided in the methodology section of this paper.

Integrating technical environmental assessment with other methods is critical for enabling innovation managers to make informed decisions to facilitate sustainable transitions within their organizations. Such transitions often require sweeping innovations in business models, products, and organizational structures. SMEs, in particular, face challenges in identifying unsustainable practices and devising strategies for change, frequently relying on tools like LCAs and Scope-3 greenhouse gas emission assessments. However, LCAs are complex and are not an exact science, as we will show in the literature review, necessitating complementary analyses and methodologies. While such integrative efforts are in their nascent stages, with initial conceptual development made in Niero et al. (2021), more empirical data is lacking. This paper focus specifically on the process of making LCAs and giving them an impact through PD.

Based on five case studies involving organizations seeking support for considering strategies for sustainable business, this paper explores various barriers and strategies to make LCA outputs actionable and effective in promoting sustainable development. These case studies provide empirical evidence and practical insights into how integration LCA with PD can facilitate innovation.

Research Question

Building on the comprehensive foundation outlined above, this article poses a critical research question: **How can LCAs, when integrated with LCM and PD methodologies, support sustainable business development and business model innovation, and what typologies can be developed to guide businesses in leveraging these tools effectively?** This research aims to investigate how the integration of LCA, LCM, and participatory design can effectively drive sustainable development and business model innovation. By analysing diverse organizational contexts, the study seeks to develop a practical typology that businesses can use to strategically apply these tools, enabling actionable sustainability improvements and fostering iterative co-design processes that contribute to both environmental and business innovation.

2 State-of-the-art

The paper combines concepts and methodologies from two diverse academic fields that are rarely combined in the literature: The combination of environmental assessments and design approaches. This state-of-the-art gives an overview of the few papers that study LCAs related to these areas and review each area separately to extract findings that can inform the current discussion. It will start with reviewing LCA as the cornerstone of the proposed framework, and continue with the relation between LCA, sustainable development, and design approaches.

Life Cycle Assessment

This section provides a succinct overview of LCA and its role in environmental impact assessment. LCA is a systematic methodology that can assess the environmental impacts throughout all stages of a product's life cycle—from raw material extraction, through production and use, to disposal. It achieves this by analyzing data on key environmental aspects, including energy consumption, greenhouse gas emissions, resource depletion, waste generation, and ecological consequences (Birat et al., 2014). LCA offers a comprehensive view of a product's environmental footprint, identifying critical areas for potential improvement to enhance sustainability. It is widely regarded as a valuable decision-making tool within business and regulatory frameworks (Birat et al., 2014; Pryshlakivsky and Searcy, 2021). The LCA framework is often described as these four separate phases: Goal and scope definition, inventory analysis, impact assessment, and interpretation (Hauschild et al., 2018).

According to Jørgensen and Forman (2009), LCA can enrich the understanding of environmental impacts associated with products and their manufacturing processes and encourages the integration of environmental guidelines into product development and the creation of cleaner, more sustainable products. Furthermore, LCA can be used to strengthen relationships between the companies in the product chain and assists companies in developing and marketing cleaner products (Jørgensen and Forman, 2009, p. 49), thus fostering sustainable development.

LCA is based on Life Cycle Thinking (LCT) and indicates that the environmental assessment goes beyond the traditional focus on production site and manufacturing processes and include impacts of a product over its entire life cycle (see e.g. <https://www.lifecycleinitiative.org/activities/what-is-life-cycle-thinking/>). LCT and LCA are increasingly recognized as essential for facilitating sustainable development, playing diverse roles in decision support. This includes providing a holistic coverage of potential environmental impacts, identifying environmental hotspots, elucidating potential trade-offs, and addressing burden shifting across different life cycle stages or impact categories (Sala et al., 2021).

An increasing range of LCA types have been developed. The two most recognized types are consequential (CLCA) and attributional LCA (ALCA) (Guinée et al., 2018). There are also a range of other types, like Backcasting LCA (BLCA), Decision LCA (DLCA), Integrated LCA (ILCA), Anticipatory LCA (NLCA) (Wender et al. 2014), Prospective LCA (PLCA), Scenario-based LCA (SLCA) (Fukushima and Hirao 2002), and probably more, all LCA types developed by independent scholars outside the ISO standards (International Organization for Standardisation 2006). So many LCA types that Guinée et al. (2018) refer to it as an alphabet soup. Guinée et al. (2018) conclude that

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these LCA types can be divided into explorative LCAs that focus on life-cycle modeling of the unknown by exploring scenarios of potential futures referred to a XLCAs, and ALCA focusing on the environmental impacts of a product system as it currently functions. However, this trajectory of research dividing the LCA world into XLCAs and ALCAs overlooks the studies not fitting this divide and hampers a constructive dialog about the use of modelling frameworks for driving innovation (Suh and Yang, 2014), like how LCAs can be used to drive sustainable development. And as concluded by Seidel (2016) many of the identified barriers indicate that the shortcomings of LCAs in positively influencing sustainable development are more related to the processes by which the LCA is incorporated into decision making, rather than to any technical issues within the LCA itself. Here we thus refrain from further dwelling on this discussion of different LCA types, but rather discuss the potential of LCA for driving sustainable development and business innovation by focusing on the challenges of making LCAs actionable in the context of organizations striving to become more sustainable. We thus do not dive into the technicalities of doing the LCA itself but focus on the input and output and the stakeholder engagement in the LCA-process.

The state-of-the-art focuses on two main challenges:

- First, the challenge of making the 'correct' LCA, which is related to the input, defining the scenario to be modelled, and what do compare it against. We will argue that it is impossible to make the precise LCA - it will always build on assumptions and delimitations (Bras-Klapwijk, 1999, p. 199).
- The second challenge is to translate the outcome of the LCA and contextualize and apply it into the organization that the LCA practitioner works in or works with, the art of LCM, so the LCA gains agency to drive sustainable development and business innovation.

The input: Defining the focus of the LCA

Defining the goal and scope of the LCA is essential and includes defining the functional unit (FU), which is a quantitative and qualitative description of the function the product system in focus delivers (Birat et al., 2014). The FU ensures that comparisons between different products or processes are made on an equal basis, reflecting their ability to perform or support the same function. The goal and scope definition delineates what is included and excluded in the LCA and should ensure that the assessment is relevant for its intended purpose.

For example, in an LCA study comparing the environmental impacts of plastic and paper cups, the FU might be defined as "the delivery of 200 ml of coffee to the consumer". This allows for a fair comparison by focusing on the equivalent function provided by different cups, regardless of material or design differences. Defining the FU precisely is crucial for the accuracy of LCA studies (Pryshlakivsky and Searcy, 2021). There are several challenges in defining a good FU. Including recycling in an LCA, for instance, lacks a consistent approach, causing confusion among practitioners (Birat et al., 2014), although there are several approaches pertaining to the reference flow to include recycling in the modelling.

When comparing a traditional disposable cup with a reusable cup defining the number of reuses and the disposal at the end of their service life is crucial for a useful

LCA. A complementary analysis like social practice perspective is a potential analysis to clearly map the scenario to be modeled helping to correctly define the FU (Niero et al., 2021). Many different complementary analyses can potentially help defining the FU depending on the product and the context.

In an LCA, a scenario can be defined as "a set of aspects describing a specific situation at a specified time" (Bisinella et al., 2021, p. 2162). Scenarios help mapping out future situations or solutions (Pesonen et al., 2000), and is a common approach to dealing with time and futures in LCA (Van Der Giesen et al., 2020). They can be used on any time horizon to evaluate the effects of epistemic uncertainty through scenario analysis, by selectively changing some aspects of the LCA model to represent alternative situations (Bisinella et al., 2021).

Bisinella et al. (2021) found that 41% of reviewed LCAs combined their LCA with additional models and modeling tools, including future scenario building. However, there is low consistency in assessing future scenarios with LCA, and studies often develop the goal and scope of future scenarios and LCA separately (Bisinella et al., 2021), leading to inconsistencies in modeling approaches and assumptions. It is important to distinguish between using scenarios to account for changes in the context of the product or service being modeled, such as a future increase in renewable energy in the electricity mix used in manufacturing, and using scenarios to define different use situations, types of users, and the impact on the FU as the product's lifetime varies.

Changes in the societal context of product systems over time is also essential as it might open for new potential scenarios, and changes in user practices and realism of modelled future scenarios are essential when doing LCAs. Understanding the user practices and their diversity in relation to a product system is crucial when defining a FU. Despite the importance of FUs to the validity of an LCA, Bisinella et al. (2021) found that only 60% of studies defined their FU, indicating the complexity and challenges in defining a FU and showing the need for complementary analyses.

The output: Translating and Contextualizing LCA Results for innovation

The LCI links the different unit processes required to deliver the products in focus in an LCA. The result of the LCI is an overview of the quantities of elementary flows in terms of resource use and emissions from the unit processes within the system boundary of the LCA. The flows are scaled to the product system through the extent to which the function of each unit process is required to deliver the FU (Hauschild et al., 2018). The next step is the Life cycle impact assessment (LCIA), which is composed of selection of environmental impact categories, classification of the flows in relation to the impact categories expressed in a common metric for the impact category, and characterization of the LCI results into aggregated indicator scores for the different impact categories. These aggregated scores constitute the characterized impact profile of the product system (Hauschild et al, 2018).

A written report summarizes the methodology, data, and findings, so they can be used for interpretation of the results and consideration of changes, which can reduce the environmental impacts of the product system. The interpretation identifies significant issues based on the results of the inventory and impact assessment, including assessment of the completeness, sensitivity, and consistency, leading to conclusions, limitations, and recommendations (Castellani et al., 2017).

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The outcome of an LCA plays a pivotal role in supporting innovation by providing innovation managers with crucial insights on how to enhance product sustainability, thereby boosting sales and reducing costs (Silva, 2012). LCA facilitates the identification of environmental hotspots within the product life cycle, directing innovation efforts towards areas with the greatest potential for impact reduction (Matthews et al., 2019). The customization of LCA approaches to meet the specific needs of R&D and business sectors ensures that the information provided is accurate and actionable (Silva, 2012). This customization helps ensure that LCA results genuinely support sustainable development and prevent misleading interpretations.

Stakeholder engagement is essential to achieve this outcome though. The integration of stakeholders in the LCA process not only yields robust results but also shifts perspectives on environmental impacts, encouraging actors to appreciate the role of design in reducing environmental impact and fostering responsible innovation practices (Yousefzadeh and Lloyd, 2021). However, a significant challenge arises in determining when to involve stakeholders and how to effectively incorporate their input into environmentally focused analyses (Berube, 2013). LCA involves numerous normative decisions, including system boundary definition, functional unit selection, impact category selection, and weighting—all of which require careful stakeholder engagement to ensure accuracy and relevance (Wender et al., 2014, p. 202).

To effectively integrate LCA with innovation practices, it is thus crucial to apply PD methods that facilitate meaningful interactions with impacted stakeholders. While ISO standards for environmental LCA recommend stakeholder engagement (Wender et al., 2014), practitioners often lack the necessary tools and expertise to implement these interactions effectively. Moreover, the transition to sustainable practices requires a comprehensive re-evaluation of innovation strategies. This transition must incorporate social and environmental considerations alongside economic factors to achieve sustainable innovation (Motta et al., 2018). Such innovation, which harmonizes economic benefits with positive social and environmental outcomes, is vital for fostering sustainable development. However, the complex, multidimensional nature of this integration poses significant challenges (Buhl et al., 2019). This complexity necessitates robust stakeholder involvement to ensure that LCA not only assesses but also effectively contributes to business innovation.

Pryshlakivsky and Searcy (2021) discuss the trade-offs between different types of LCA for decision-making purposes, using examples on one hand of streamlined LCAs that reduce complexity to ease decision-making at the cost of depth of analysis and potentially neglecting externalities, and on the other hand LCA expansion that increases complexity to increase scientific credibility, but with trade-offs in reliability for relevance. The needs of potential decision-makers from an LCA, including the level of complexity and specific impact categories of relevance, might be unknown to the LCA practitioner (Birat, 2023). Proper stakeholder involvement and integrating LCA with complementary tools and analyses can better facilitate robust and positive decision-making, as argued by Sevigné-Itoiz et al. (2021). This is also supported in older LCA research by Bras-Klapwijk (1999) that points out that actors might be in a constant discursive struggle about the nature of the environmental problem as different actors have different vested interests. Broad participation of actors in an LCA might minimize political struggles around the LCA and increase the possibilities for the results of the LCA to be used.

Bisinella et al. (2021) emphasize that the interpretation phase can be strengthened by using additional models in collaboration with stakeholders, such as Multi-Criteria Decision Analysis approaches. Other research also argues for increased stakeholder involvement and considerations of societal needs not to undermine the value and credibility of an LCA (Notarnicola et al., 2017; Sevigné-Itoiz et al., 2021).

In summary, there is a gap in the literature about how individual organizational needs can be integrated into the goal and scoping of an LCA and the strategies for incorporating LCA results. Additionally, further investigation is required to determine how different LCA types can be applied without reducing usability. Businesses are encouraged to view LCA not just as a compliance tool but as a strategic framework for long-term sustainability planning (Clune and Lockrey, 2014). The effective use of LCA in design often requires collaboration between different fields, including social sciences and engineering, to address complex sustainability challenges (Clune and Lockrey, 2014), which we illustrate in this paper by combining LCA and PD.

Complexity and Limitations

Limitations of Objectivity in LCA: Some literature also emphasizes that LCAs are inherently based on assumptions and boundaries (Bras-Klapwijk, 1999), which prevent them from being fully objective or exhaustive. Normative and often implicit assumptions cannot be made in an objective and decisive way and highlights the risk of premature 'closure' of LCA analyses.

Normative Assumptions and Knowledge Gaps: Simplifications are often made in LCAs for economic and practical reasons, i.e. what cannot be calculated is not included, which means certain impacts, components, or alternatives may not be fully addressed. This can lead to oversights in decision-making processes. Stakeholders need to understand these limitations to interpret LCA results effectively.

The Value of Stakeholder Engagement and Complementary Analyses: The need for broad participation in LCA to minimize biases and address diverse interests and interpretations is essential.

Design Approaches and Life Cycle Assessment

Design Thinking and PD stand out as pivotal methodologies for fostering sustainable innovation. These approaches incorporate collaborative and iterative processes, actively involving multiple stakeholders in dialogue, idea generation, and problem-solving. To facilitate these processes effectively, tools such as design games, prototypes, and co-design frameworks are utilized, which are well-documented by Kjaersgaard et al. (2021) and Brandt (2006). The landscape of design methodologies is vast, comprising Design Thinking, Participatory Design, Double Diamond, User-Centered Design, Service Design, and Co-Design, each equipped with distinct tools and methods aimed at transforming the status quo as highlighted by (Clausen et al., 2020).

A unique and innovative aspect of the Staging approach in PD is the combination of the action-oriented methods from the Scandinavian tradition of participatory design and innovation with the strategic insights from Science and Technology Studies (STS). This blend provides a multifaceted view on change and innovation, essential for navigating the complexities inherent in sustainable development. Over many years, these integrated design methodologies have evolved significantly at institutions like the Technical

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University of Denmark and Aalborg University within their respective Design and Innovation and Sustainable Design programs (Andreasen, 2011; Clausen et al., 2020). As Andreasen (2011) articulates, this melding of design and engineering creates a unique scientific field characterized by diverse research paradigms.

However, research on integrating LCA with these robust design approaches remains sparse. Traditionally, LCA has been more commonly associated with eco-innovation rather than broader sustainable development (Motta et al., 2018). Even though integrating LCA into innovation processes has been shown to potentially enhance environmental sustainability, only few instances of LCAs facilitating innovation has been observed (Motta et al., 2018). A targeted literature search for articles combining LCA with various design methodologies yielded only 11 relevant studies in Scopus, with Clune and Lockey (2014) being the only paper that explicitly integrates LCA with design methods.

Clune and Lockey (2014) effectively demonstrates how the integration of LCA within the Double Diamond framework can pinpoint high-impact areas, thereby facilitating the development of actionable sustainability strategies. This alignment showcases the potential of LCA to drive sustainable development by catering to specific contextual needs.

Reviewing literature that cites Clune and Lockey (2014) over the past decade reveals a limited engagement with this integrative approach. Among 59 papers, only four directly relate to LCA, with two linking LCA with design methods. These papers reiterate the notion that LCA should be able to drive sustainable dvelopment. For example, De Laurentiis et al. (2016) discuss LCA's role in providing an evidence base for transitional strategies, and Teah et al. (2015) consider the use of LCA in supporting decision-making for a more cost-effective alternative to systematic field experiments or laboratory research. Borrion et al. (2019) explores how simplified LCA tools can help non-experts make informed decisions, particularly through a co-design process in city infrastructure planning, making LCA a foundational tool within the design process to facilitate sustainability discussions.

Furthermore, the applicability of LCA in design processes extends to other domains. Boix Rodríguez et al. (2021) integrate LCA with design methods in the engineering of face masks, demonstrating how it can guide engineers in making informed decisions throughout the design process. This integration, although straightforward and lacking participant engagement in the face-mask case, illustrates the versatility of LCA in supporting informed design choices.

Despite these exploratory efforts, no study significantly advances the foundational findings set by Clune and Lockey (2014), which detailed how the iterative nature of the Double Diamond method identified areas within elder care that had the highest environmental impact, linking these to day-to-day practices and organizational behaviors, and exploring alternative practices that suggested strategies potentially reducing carbon emissions by 54% over a decade.

In conclusion, while the integration of LCA with design methods shows promise, substantial work is needed to apply these insights more broadly and effectively across different empirical fields. As shown by Clune and Lockey (2014) this integration not only enhances the capability to tailor business innovation to specific organizational needs but also leverages the strengths of engineering, social sciences, and design to create impactful environmental solutions. Developing general methods that contextualize LCA knowledge within design processes, rather than creating new tools for each project like Borrion et al. (2019), represents a critical area for future research and application.

Methodology

The design methodology presented here has been developed over several years through our work in the Sustainable Design master's program in engineering at Aalborg University. This methodology has evolved significantly based on the project module on the design of product-service systems and the course on lifecycle-based environmental assessment of products and systems offered in the 4th semester. These educational components have laid a robust foundation for iteratively combining various analytical approaches and methods from STS and Design, and techniques for conducting comprehensive environmental assessments. STS and Design have for many years been integrated in the staging approach (Clausen 2020). The novelty in this paper is the integration with LCA.

Engagement and Operationalization through PD and the Staging Approach

Our methodology leverages participatory design methods to actively involve stakeholders in the innovation process (Clausen et al., 2020; Pedersen, 2020). In the staging approach the designer or innovator orchestrates interactions among stakeholders using various design artefacts, such as prototypes, digital simulations, texts, visualizations, etc. The students carry out the project based on a double diamond model (Clune and Lockrey, 2014) that they and adapt to their projects. The double diamond is integrated with the problem-based learning model of AAU, as it iterates between discover or exploring the case and defining the problem, usually through several iterations, before starting iterations of developing a solution and delivering it to the case partner. This methodology facilitates open dialogue, idea generation, and collaborative problem-solving, aligning the innovation process with real-world needs and expectations.

There are thus three core aspects of our staging approach, 1) the overall design process and research design of the project, 2) the interaction with the stakeholders, 3) and the objects designed to facilitate the interaction and translate the outcome into relevant context.

The process: In the overall framework of the double diamond the discover-define iterations relate to the input section in the state-of-the-art, defining the focus of the LCA, and the developing-delivering relates to the output and the work with translating the LCA-outcome to something useful for the case partner.

The objects: The design artefacts made throughout the project are instrumental in visualizing solutions and exploring different scenarios, making the abstract aspects of design and impact assessment tangible and comprehensible.

The stakeholder engagement: By facilitating interactions through these objects, the staging approach enables stakeholders to delve into diverse perspectives and co-create solutions that are not only innovative but also sustainable.

This method is particularly effective in fostering a participative environment where all voices are heard, and collective intelligence is harnessed to address complex challenges. For further specifics on the participatory design methods employed, we refer to Clausen et al. (2020).

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Integrating Environmental Assessments with LCA

A key component of the proposed methodology is the Life Cycle Assessment (LCA), sometimes combined with other analyses as necessary. Conducted according to ISO 14044:2006(E) standards, the LCAs utilizes SimaPro software with data sourced from the Ecoinvent 3 database. The cases apply the ReCiPe 2016 Midpoint (H) method to assess environmental impacts, ensuring that sustainability evaluations are both rigorous and compliant with international standards.

Application in Real-World Case Studies

Applying this comprehensive methodology in case studies involves detailed documentation of how participatory design and environmental assessments converge to influence sustainable development. Each case study illustrates how these elements interact to facilitate innovation, detailing the processes, challenges, and achievements. Insights gleaned from these case studies will be instrumental in refining the methodology and enhancing its applicability across different sectors and projects.

Empirical basis

The paper is based on a range of case studies conducted by students in Sustainable Design, a Master of Science in engineering at Aalborg University, the Technical Faculty of IT and Design, the Institute for Sustainability and Planning. The cases studies are conducted on the 4th semester of the bachelor where the focus is on circular economy, life-cycle thinking and assessments, product service systems, and related concepts. The students make a 10-ECTS semester project and 5-ECTS LCA analysis in groups of 4-6 students, corresponding to approximately 1600-2400 workhours. Aalborg University is based on a pedagogy of problem-based learning (PBL) that is project-based and where most teaching takes places through project supervision. The projects have had supervision meetings with one of the authors at least once every 1-2 weeks during the project period and have also an LCA-supervisor attached. All projects have been evaluated in depth and their validity accessed through oral exams of 3-4 hours per project, a very extensive oral-exam format, and evaluation with external partners done subsequently to evaluate the usefulness of the project outcomes.

The research interest for this paper have developed over a range of years, and semester projects have run within the current framework for 5-6 years. Projects have been set up the last 2 years employing the framework discussed in this paper with the intent to gather empirical data of relevance for this article. Not all student projects were of high enough quality, or the students choose other frameworks as they were allowed to do within the PBL framework, and where thus discarded. The five most interesting projects were chosen as exemplary cases, of which two are unfolded the remaining four summarized.

Results: From Analysis to Innovation

Municipality

A large and relatively rich urban municipality with 75.000+ citizens. An employee responsible for sustainable transition in the municipality approached the university to collaborate with students on identifying potentials for reducing environmental impacts of the municipal activities, which could be anything from the municipality's use of furniture to food, and electronics, a very wide scope.

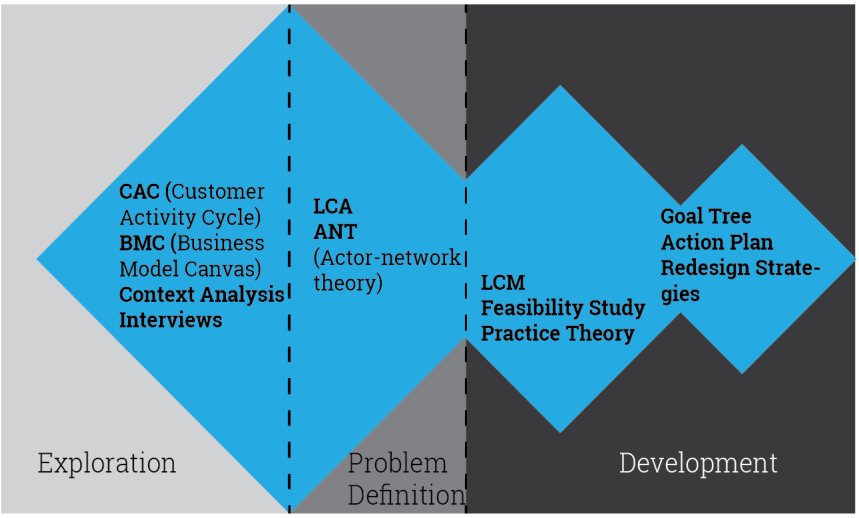


Figure 1 - illustration of the double diamond inspired design process including the various analyses done at different stages

In the exploration phase (see Figure 1), the project focus was refined to IT equipment with the municipality's input. A variety of complementary analyses were carried out throughout the project. The project adopted a triple-diamond in contrast to the traditional double-diamond. The first phase concentrated on scoping the project and defining the LCA. The second phase involved engaging stakeholders through design games and interviews to explore sustainable development options based on the LCA outcomes. After identifying five core issues, the project narrowed down to one specific problem. The final phase developed multiple solutions, ultimately converging on a final design recommendation. Table 1 summarizes stakeholder engagement throughout the project.

Table 1 - Stakeholder engagement overview

Date	Type of Engagement	Stakeholder	Category of Meeting
Feb	Interview	Case Holder	Meeting with Case Holder
Feb	Design Game, Business Model Canvas (BMC)	Case Holder	Meeting with Case Holder
Feb	Interview	Procurement and Contracts	Meeting with Town Hall Staff

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Feb	Interview	IT Department	Meeting with Town Hall Staff
Mar	Interview	Case Holder	Meeting with Case Holder
Mar	Interview	Finance Department	Meeting with Town Hall Staff
Mar	Interview	Knowledge Sharing	Meeting with Case Holder
Apr	Interview	Climate Network	Meeting with Town Hall Staff
Apr	Interview	KMT	Meeting with External
Apr	Interview, Locked Devices Discussion	Case Holder	Meeting with Case Holder
Apr	Interview	Education Department	Meeting with Case Holder
Apr	Interview	Supplier	Meeting with External
May	Design Game, Feasibility Study	Case Holder	Meeting with Case Holder
May	Interview	Procurement	Meeting with External
Jun	Design Game, Goal Setting	Case Holder	Meeting with Case Holder
Jun	Interview	COMM2IG	Meeting with External

As visible from Table 1, the first diamond of the project took 3 months, as understanding the organization, the problem, and iteratively defining and developing the LCA is the most time-consuming part of the project. As pictured in Table 2 below, at the outset and during the project a whole range of unanswered questions emerged, illustrating the necessity of continuous stakeholder engagement.

Table 2 - Open and emerging questions

Category	Open Questions at the Outset
Scope Definition	Which areas or products within the municipality's activities should be prioritized for environmental impact reduction?
LCA Focus	How can LCA be applied effectively within a municipal context to drive sustainability?
Stakeholder Engagement	Which stakeholders need to be involved to obtain a comprehensive view of sustainability practices in the municipality in relation to the identified activities?
Product-Specific Impacts	What environmental impacts are associated with the municipality's IT equipment?
Lifecycle of IT Equipment	How long are laptops used before disposal, and what are the reasons for their replacement?
Waste Reduction	How can the municipality reduce waste from prematurely discarded IT equipment?
Sustainable Procurement	What are the possibilities and limitations of sustainable procurement practices within the municipality?

Operational Practices	How do current administrative practices impact the environmental footprint of municipal operations?
Legislative Compatibility	How do legal requirements influence the municipality's ability to manage and dispose of IT assets sustainably?
Environmental Hotspots	Which specific practices within IT equipment management have the largest environmental impact?
Laptop Lifecycle Optimization	How can the lifecycle of laptops be extended to reduce environmental impact?
Stakeholder Practices	What are the key organizational and operational barriers to adopting more sustainable IT practices?
Behavioural Patterns	How do employee behaviours, such as logging off improperly, lead to functional laptops being disposed of unnecessarily?
Warranty and Replacement Policies	How does the expiration of warranties influence the premature disposal of laptops, and what alternative policies could reduce this impact?
Departmental Ownership	Why are inactive or "dead" laptops not shared across departments, and how can this be improved to avoid waste?
Impact of Bulk Purchases	How do bulk purchases by the IT department contribute to premature replacements, and what changes could reduce waste from over-procurement?
Tracking and Management System	What system could effectively track the lifecycle of laptops, from purchase to disposal, to improve decision-making?
Engagement Methods	Which participatory design (PD) and Life Cycle Management (LCM) methods are most effective for engaging municipal stakeholders and aligning them with sustainability goals?
Redesign Strategies	What specific redesign strategies could be implemented to make IT procurement and usage more sustainable?
Feasibility of Solutions	Are the proposed changes in procurement and management practices feasible within the municipality's existing structure and budget?
Alignment with Goals	How can the LCA results be translated into actionable steps that align with the municipality's sustainability objectives?
Prioritization of Impact Categories	Which environmental impact categories (e.g., land use, global warming potential) are most relevant for the municipality to focus on, and how do they align with broader sustainability goals?
Stakeholder Buy-in	How can stakeholders be continuously engaged to support long-term adoption of sustainable practices?
Quantifying Impact Reduction	To what extent can the suggested changes reduce the municipality's overall emissions and resource use?

This paper was presented at ISPIM Connects Osaka – Connecting and Empowering Society, Osaka, Japan on 2-4 December 2024.
Event Proceedings: ISBN 978-952-65069-7-5

Process: Defining the Focus of the LCA and Identifying Key Issues

The project scope was refined to focus on IT equipment, specifically laptops used by over 500 administrative employees at the town hall. Insights into daily practices were gathered through interviews with employees across departments, complemented by informal conversations and observations to deepen understanding of IT equipment management. Quantitative data, including procurement records and climate reports, were analysed to substantiate the findings.

Initial analyses mapped out the laptops in terms of quantity, flow, procurement details, and reasons for becoming inactive or obsolete. In the defining phase, the team identified key issues that shaped the LCA's focus, revealing that managing laptop lifecycles—particularly addressing premature disposal driven by administrative practices and warranty constraints—was a critical area. This iterative process led to the central question: How could the municipality extend laptop lifespans while reducing environmental impact? The five key issues identified were:

- Functional laptops are sent for disposal without being reused because employees do not log out correctly, locking the devices due to encryption.
- Functional laptops are often replaced once their warranty expires, typically between 3 to 5 years, leading to premature disposal despite their potential for longer use.
- Laptops that are no longer in use (‘dead units’) are not shared across departments due to economic ownership issues, resulting in wasted resources.
- Bulk purchases by the IT department lead to premature replacements, as departments receive new laptops for free, and these laptops might start their warranty period before actual use.
- There is a lack of a comprehensive system to track the flow, purchase reasons, on-boarding, maintenance, and replacement of laptops, leading to inefficiencies and oversight in laptop management.

The question was which problem to focus on and following that how to define the FU and scenarios to be modelled in the LCA. As illustrated by Figure 1, Table 1, and Table 2, a lot of work and stakeholder engagement have already been necessary at this point, and the project has only arrived at the point of defining the FU and Scenarios at this stage. Besides identifying laptops as a good candidate for reducing the environmental impact of the Municipality and pinpointing the environmental hotspots related to laptops, the question is which problems realistically can be handled by the Municipality, which led to another stakeholder engagement to weigh the key issues against different criteria, see Table 3, which led the project to choose key issue 2. Different FUs were considered, eventually defining the FU as "supporting 500 employees to perform office tasks over 10 years using a laptop." Several scenarios were explored, including extending laptop use beyond the warranty period, implementing regular maintenance routines, and redistributing laptops within the municipality to avoid unnecessary replacements. These scenario models were key in illustrating potential reductions in environmental impact for different models, and potentially instrumental in guiding decision-making. For this case,

these first three months that led to the FU were the most important, interesting and illustrative.

Table 3 - Feasibility weights

	Key issue				
	1.	2.	3.	4.	5.
Knowledge Potential	+	0	+	-	0
Environmental Potential	+	+	0	0	+
Caseholder's Interest	+	+	0	-	0
Students Interest	0	+	+	-	0
LCA	+	+	+	0	-
Time to complete	+	+	+	-	+
Score	5	5	4	-4	1

The final co-developed scenario demonstrated the potential for significantly reducing the environmental footprint by extending laptop lifespan through better management and maintenance practices. The innovative suggestion was adapting the idea of a produce-service system to the internal organisational setting of the municipality, removing the specific department budgets from the equation when buying laptops, moving the ownership of the laptops to the internal IT service department. The departments thus only pay for the service of having a certain number of functional laptops, and the IT department with the necessary competence, knowledge, and overview can then optimise the system. The upper management of the municipality were presented the outcome and chose to adopt the suggestions and start a discussion of implementation.

Learnings from Integrating the Staging Approach with LCA in the Municipality Case

The Municipality case illustrates the value of combining a design-oriented staging approach with LCA to drive actionable, sustainable change.

- **Visualization and Scenario Exploration**: Lifecycle diagrams and scenario models, visually connect to organisational diagrams and practices, clarified the complex sustainability dynamics of IT management, allowing stakeholders to see how organizational practices impact laptop lifespan and environmental effects. Visualizing issues like premature disposal highlighted strategic opportunities, such as extending laptop use beyond warranties and centralizing IT ownership.
- **Stakeholder Engagement and Co-Creation**: Iterative engagement with administrative and IT staff enabled stakeholders to shift from data providers to solution co-creators in the project. This process identified factors influencing laptop lifespan (e.g., warranty policies, logout issues), grounding LCA scenarios in real organizational practices, and fostering acceptance of proposed solutions. Design games are here essential to enable stakeholders to understand the ongoing analysis and entice them into interactions.
- **Outcome and Key Insights**: Using the staging approach, the project translated LCA insights into actionable recommendations, including a central IT ownership model that supports laptop lifespan optimization. This service-based model aligns with

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municipal sustainability goals, showcasing how staging can bridge LCA findings with practical organizational implementation.

Interestingly, the students driving the project saw the degree of co-design as low, as there were few interactions in the last diamond phases where the solutions were developed. However, in this case the crucial period was in the first diamond phase where the problem to be solved was defined, due to the vague and undefined nature of the initial problem.

Furniture rental

This case study focuses on a furniture rental company in a medium-sized Scandinavian city, referred to as "Furniture." The company aimed to scientifically document the sustainability of its business model to support marketing efforts. Their business model promotes reduced environmental impact and increased convenience as key selling points. The project also involved a sustainability consultancy that cooperated with Furniture. The case focused on making an in-depth LCA to identify areas for potential environmental impact reductions in their furniture management process. The study specifically targeted furniture used by exchange students, analysing aspects from procurement through to end-of-life processes, using both qualitative and quantitative data.

Process: LCA Focus and Key Issues

Unlike the Municipality case, the scope was pre-defined as furniture for exchange students, allowing the project to follow a traditional double-diamond process. The primary LCA focused on a basic furniture package of a chair, desk, and bed, though the FU and comparison scenarios were open questions. Later, the FU was defined as fulfilling the furniture needs of 100 students, allowing for meaningful scenario comparisons between new, rented, and second-hand furniture. Table 4 shows a summary of the initial open questions and those that emerged throughout the project. In comparison with the Municipality case, it is apparent that this case is much simpler with the radically smaller number of open question identified by the students.

Table 4 - open and emerging questions for the Furniture case

Category	**Question**	
LCA Application	How can Life Cycle Assessment (LCA) be effectively applied within a furniture context to assess environmental sustainability?	
Data Gaps	Are there any missing data points (e.g., exact materials used, production methods) necessary for a complete LCA?	
Product-Specific Impacts	What are the main environmental impacts associated with different types of furniture?	
End-of-Life Management	What are the current disposal and recycling practices, and how effective are they?	
Hotspot Identification	Which stages in the furniture lifecycle (e.g., production, usage, disposal) have the largest environmental impact?	
Stakeholder Engagement	Which stakeholders (e.g., suppliers, users, maintenance teams)	

	need to be involved to obtain a comprehensive view of the furniture's lifecycle?
Lifecycle Optimization	How can the furniture lifecycle be extended to reduce environmental impact?
Product Requirements	Are there specific requirements for the type of furniture procured to ensure its lifespan can be extended?
Maintenance Practices	How do current maintenance practices affect the longevity and sustainability of furniture items?
Procurement Practices	What sustainable procurement practices are feasible within the organization?
Reuse and Repurposing	What are the possibilities for reusing or repurposing furniture within the organization?
Scenario Modelling	What scenarios (e.g., increased maintenance frequency, alternative logitiscs, reuse models) show the most potential for impact reduction?

The initial analysis involved mapping the life cycle of furniture—from procurement to disposal—examining procurement methods, usage duration, replacement, and disposal practices. Interviews with external stakeholders, including suppliers and disposal teams, provided insights into the furniture lifecycle. Key issues identified included premature disposal of functional furniture due to minor, repairable damage, frequent replacement cycles driven by aesthetics rather than function, and a lack of standardized tracking for furniture condition, leading to operational inefficiencies and increased environmental impacts.

Scenario Development and Empirical Research

Furniture assumed its model replaced the need for virgin furniture purchases and disposal but lacked data on actual customer practices. The student team conducted an empirical study with 94 exchange students, revealing three main usage patterns: 83% bought second-hand and donated it after use, 10% bought new and donated after use, and 7% bought new and disposed of furniture after use. These findings informed the LCA baseline scenario, against which Furniture's impact was compared.

Figure 2 presents a comparison of environmental impacts across the scenarios. The baseline scenario showed that Furniture's model reduced environmental impact by approximately a fourth compared to the alternatives it substitutes, though it was less impactful than initially assumed. The LCA also highlighted transportation between storage and user as a major contributor to environmental impact, particularly if Furniture would expand to neighbouring cities, which would require decentralized storage for a more sustainable distribution.

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Event Proceedings: ISBN 978-952-65069-7-5

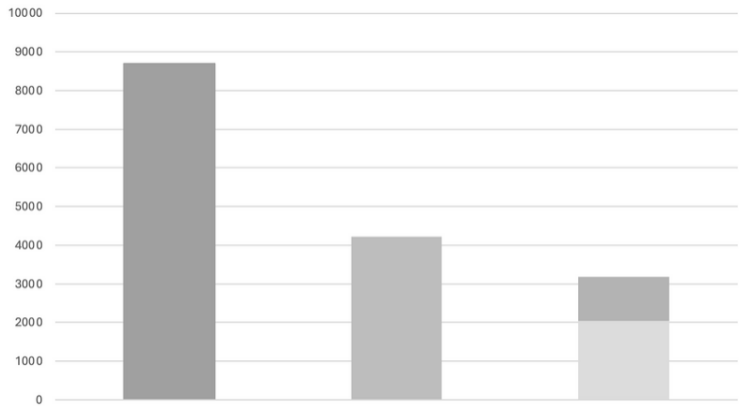


Figure 2 - environmental impact scenario comparison in kg CO2 eq

More importantly for the development of the company, the different aspects of the business model could be separated in the LCA, showing what parts of the operations have the biggest environmental impact, which is important input for considerations about LCM. Transport had the biggest impact in Furniture business model. A sensitivity analysis compared their operations against the other practices of the students in case Furniture expanded to neighbouring cities, as they considered.

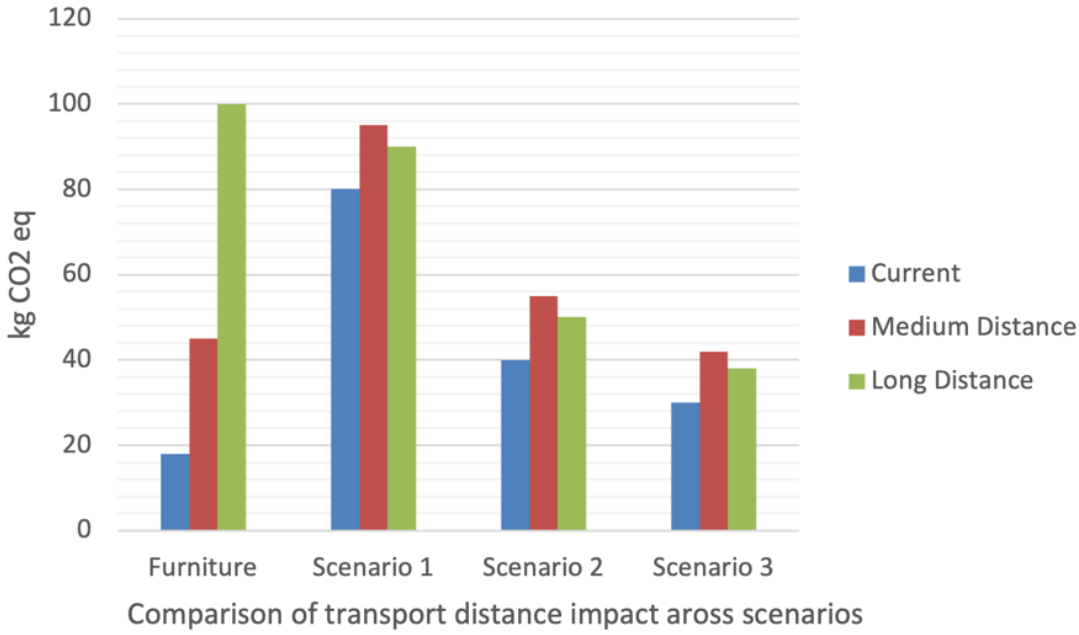


Figure 3 – comparison of different business scenarios across varying distances

It showed Furniture becoming less sustainable than the alternatives, scenario 1-3, as transportation distances increase. Expansion would thus require, as mentioned, decentralized storage to reduce transportation distance.

Action Plan and Business Model Innovations

Based on the LCA findings, the team went into LCM-considerations and developed an action plan prioritizing reduced transportation through logistics improvements, such as consolidated deliveries, route optimization, and partnerships with decentralized storage facilities near student hubs. The engagement process revealed that “land use”, a specific impact category in the database used in the SimaPro software, was a socially significant impact category alongside global warming potential in this cultural context, i.e. preserving forest and nature areas were highly valued, offering a marketing angle for Furniture.

Stakeholder engagement was crucial, especially with existing and potential users, to accurately represent the practices Furniture's business model replaces. Internal stakeholder engagement also ensured that Furniture staff understood the critical role of logistics in their environmental footprint.

The positive reception from the consultancy's LCA specialists validated the project's methodological approach, particularly the use of empirical data to define scenarios. Ultimately, this project highlighted the importance of scenario modelling informed by real-world practices for accurate LCA, and it steered Furniture towards a more sustainable business model.

Case Study: Office – Optimizing Circular Economy in Office Furniture Rental

Office, a company specializing in office furniture rental and refurbishment, operates within a competitive market traditionally dominated by linear economic practices, where products are typically discarded after use instead of being recycled or reused. Office distinguishes itself by providing office furniture rental services that focus on maintaining, refurbishing, and circulating furniture across different clients. This circular economy model aims to extend the furniture lifecycle, reduce waste, and minimize resource consumption, contributing to global sustainability goals.

Problem Identification

The collaboration with Office began with an exploration of specific sustainability challenges in their operations. The primary objective was to assess the environmental impacts of Office’s current Product-Service System (PSS) and identify opportunities for improvement. Key challenges centered around lifecycle management of office furniture and incorporating stakeholder feedback to enhance environmental performance.

Life Cycle Assessment (LCA)

An LCA was conducted to quantify the environmental impacts associated with Office’s furniture, from production to disposal. This analysis identified critical impact points or “hotspots” where environmental improvements could be prioritized. Results suggested that Office’s circular model has the potential to reduce impacts compared to a linear model, especially if furniture lifespan is extended significantly. Specifically, Office’s furniture would need to last more than twice as long as furniture in a linear model, especially if linear consumers recycle furniture after initial use. Initially Office, like Furniture, had only compared against a worst-case scenario that did not seem realistic.

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Event Proceedings: ISBN 978-952-65069-7-5

Sensitivity analysis further highlighted that furniture lifespan and disposal methods were particularly influential in determining the system's emissions. However, as a young company, Office lacks verified data on actual furniture lifespan and required levels of service, making these assumptions provisional.

Key Issues

Through LCA and stakeholder engagement, three primary issues were identified:

1. **Lack of Data and Information**: Insufficient data on damage history, repair needs, and patterns limits Office's ability to optimize furniture maintenance and lifecycle management.
2. **Weak Customer-Furniture Connection**: Customers often lack a sense of attachment to the rented furniture, leading to less careful handling and increased wear and tear.
3. **Transportation and Damage**: Repeated transport between clients raises the risk of damage, potentially shortening furniture lifespan and increasing environmental impact.

The project focused on addressing the first two issues, recognizing that resolving them could significantly improve sustainability outcomes.

Proposed Redesign Strategies

To address the identified issues, the project proposed two design strategies:

- **Design for Standardization**: This strategy aims to establish a data collection standard for tracking each piece of furniture's damage history, repairs, and recurring issues. Clear guidelines for data collection would ensure that this process is intuitive and accessible for both Office and its customers, enabling a more effective and consistent approach to furniture lifecycle management.
- **Design for Product Attachment**: This approach seeks to foster a stronger connection between customers and the rented furniture, encouraging more careful handling and ultimately prolonging furniture lifespan. Creating a sense of emotional and psychological attachment to the product has the potential to enhance its durability and reduce wear.

Conclusion

The project concluded that implementing a platform where customers can report damages, detailed repairs can be documented, and data can be collected systematically to support the long-term viability of their circular model would be the most promising solution. While the platform's role in fostering customer attachment to the furniture remains uncertain, it shows potential for enhancing product care. A co-design process involving Office and its stakeholders is recommended to further develop this platform and explore ways to strengthen the customer-furniture connection, thereby supporting Office's sustainability objectives.

Coffee mugs

This case study examines the environmental implications of replacing single-use paper cups with reusable coffee mugs in a café setting. Like the furniture case, it was crucial to understand the user practice to define realistic scenarios to model. The FU was defined as eight servings of 250 mL hot liquid at temperatures between 70-100 degrees Celsius, reflecting the expected lifetime of the new reusable mugs. However, the return rate, critical to estimating the number of mugs needed for the FU, was critical to estimate and qualify.

1. **Scenario Complexity and Return Rate Estimation:** While the FU was straightforward, defining realistic scenarios required detailed empirical data, particularly on consumer practice, to accurately estimate the number of reusable mugs needed per FU. This highlights the importance of empirical scenario modelling for actionable LCA insights
2. **Environmental Impact of Production and Sanitization**: While reusable mugs performed better in most environmental categories, high impacts on climate change and fossil resource use reveal the need to consider lifecycle impacts, particularly in production processes reliant on virgin materials and energy-intensive methods.
3. **Infrastructure Limitations and Agency Constraints**: The company's limited control over system design—such as return infrastructure and sanitization—illustrates challenges in implementing sustainable innovations without broader system changes, a common theme in aligning LCA with actionable outcomes.
4. **Design Challenges in User Engagement**: Shaping user practices around return rates emerged as the main challenge since changes in cup design or return infrastructure were unfeasible.
5. **Sustainability vs. Practical Feasibility**: The project concluded that while reusable mugs are theoretically sustainable, achieving the high (80%+) return rate of the cups for a sustainable impact may be difficult. This finding emphasizes the need for grounded scenario modelling to inform realistic, context-sensitive sustainable solutions.

The case itself was straightforward. The LCA was needed to estimate the necessary success rate, for the business model to be sustainable. The LCA also revealed where the environmental impact lay, where in the value chain and which impact categories, which the students engaged the stakeholders around, to identify the viable design actions by the company.

Concrete tiles

This case study examines a project focused on designing sustainable concrete tiles, referred to as "Concrete." The project aimed to integrate circular economy principles into Concrete's business model, enhancing sustainability through the collection, reuse, and lifecycle optimization of pavement tiles. Key objectives included establishing a profitable model for tile reuse.

This paper was presented at ISPIM Connects Osaka – Connecting and Empowering Society, Osaka, Japan on 2-4 December 2024.
Event Proceedings: ISBN 978-952-65069-7-5

1. **Goal and Scope Definition**: The LCA aimed to identify and mitigate environmental impacts across three tile types, with the FU set at 1 m² of sidewalk surface over 20 years.
2. **Key Findings**: The project identified a significant issue with the premature disposal of tiles due to minor damages, often resulting from improper handling and limited maintenance skills. Cement production contributed approximately 80% of the tiles' climate impact, followed by transportation and recycling processes, directing efforts towards reducing cement content and enhancing tile reuse potential.
3. **Stakeholder Engagement**: Engaging diverse stakeholders, including municipal staff, contractors, and decision-makers, uncovered perspectives on tile function, aesthetics, and maintenance that impacted tile lifespans and environmental outcomes. Municipal workers often viewed tiles mainly as functional infrastructure, while residents and decision-makers emphasize aesthetic qualities.
4. **Proposed Solutions**: A significant barrier to sustainability was the lack of coordination and shared understanding among different actors. By mapping these relationships, the project could propose more integrated and collaborative approaches to tile management. The project suggested a multi-pronged approach, including enhanced maintenance training and stakeholder alignment to prevent tile damage and premature disposal, supported by collaborative tile management strategies.
5. **Concluding Insights**: By integrating circular economy principles and stakeholder-driven improvements, the project provided a roadmap for sustainable tile management, demonstrating the concrete industry's potential for reduced environmental impact.

By addressing both behavioural and systemic issues, the project aimed to align stakeholder interests, promote sustainable practices, and stabilize long-term commitment to a circular approach in Concrete's operations.

General findings

The cases underscore the complexities involved in defining FUs and scenarios and comparing them with potential future scenarios to support LCM decision making. There is little novelty in finding that defining FUs can be challenging. The novelty lies in how the combination with stakeholder engagement facilitated by design methods can meet this challenge, which we delve into in this sub-section. Stakeholder engagement play a critical role in ensuring that the scenarios modelled in LCAs are grounded in actual practices, and in co-analysing the outcome to translate it into the organisation and facilitate changes to the business model through considerations about LCM for sustainable development, based on Garcia Sanchez et al (2004). Some typical challenges encountered in the analysed cases:

- **Dynamic Scope Definition and Stakeholder Engagement**: Early and continuous stakeholder engagement is essential for accurately defining the scope and focus of the LCA. Engaging diverse stakeholders, such as administrative staff in Municipality and users in Furniture, helped refine project objectives and identify relevant environmental hotspots, ensuring that the analyses addressed the most impactful and actionable areas.

- **Empirically Informed Scenario Modeling**: Empirical data collection on existing practices (e.g., laptop usage patterns or furniture handling by students) proved vital for establishing realistic baselines and functional units. This approach enabled the development of credible comparative scenarios, enhancing the accuracy and relevance of LCA outcomes. For example, the baseline scenario in the Furniture case corrected the company's initial assumptions, providing a more realistic impact assessment.
- **Visualization as a Tool for Engagement and Insight**: Design artifacts, such as lifecycle diagrams and scenario models, were crucial in simplifying complex LCA findings, making them accessible and actionable for stakeholders. These visualizations facilitated a clear understanding of the impacts of current practices and potential improvements, bridging the gap between abstract analysis and concrete decision-making.
- **Iterative Co-Creation for Solution Development**: In both cases, the iterative engagement process allowed stakeholders to shift from passive data providers to active co-creators of solutions. By involving stakeholders in exploring and refining LCA scenarios, the projects fostered buy-in and enhanced the likelihood of successful implementation, as seen with the IT service-based ownership model in Municipality and the transportation-focused logistics improvements in Furniture.
- **Linking LCA Insights to Business Model Innovation**: The integration of LCA with LCM revealed specific areas for business model innovation. In Municipality, the shift to a centralized IT ownership model addressed key environmental challenges, while Furniture's LCA insights led to logistical optimizations. This approach illustrates the value of using LCA findings to inform strategic decisions that align environmental sustainability with operational goals.

This approach points to two core issues when conducting LCAs: the specificity of the analysis and the necessity of stakeholder engagement. Addressing these challenges of use of LCA's requires careful attention to both dimensions.

The first issue is the **specificity of the LCA**, which relates to how well-defined the FU and scenarios are in the analysis. For example, in the **Furniture** case, the products being modelled, such as beds, desks, and chairs, were straightforward, as were the products in the **Tiles**, **Office**, and **Mugs** cases, where the business models clearly defined the products. However, in the **Municipality** case, determining which product group would yield the greatest environmental impact required a more exploratory approach and entailed a high engagement. This aligns with Clune and Lockrey's (2014) eldercare study, where the products and impacts were less defined, requiring more open-ended analysis. However, the **Design** case that also involved low specificity, as the focus was on conducting an initial hot-spot analysis rather than achieving a highly targeted LCA, entailed a low engagement. Here, the goal was to lay the foundation for more specific innovation activities subsequently, that would then require a deeper engagement.

The second issue focus on the **depth of stakeholder engagement** during the LCA process, which is critical for ensuring that the results are actionable and aligned with stakeholder needs. Effective engagement helps stakeholders understand and agree with the problem framing and scenarios, draws on their knowledge through co-analysis, and facilitating a better reception and use of the outcomes. In the **Furniture** case, even though the products were specific, the initial FU and scenarios were misaligned, making stakeholder engagement essential for correcting these errors. Similarly, in the

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Municipality case, the low specificity of the analysis demanded high stakeholder involvement to iteratively define the scope of the LCA. In contrast, the **Design** case required less engagement, which turned out to be appropriate since the primary goal was to perform an initial assessment, not to dive into detailed stakeholder-driven scenarios.

These empirical examples illustrate the need for a balanced approach that considers both the specificity of the LCA and the depth of stakeholder engagement, ensuring that the analysis is both relevant and actionable for fostering sustainable development.

Discussion: Design and Stakeholder Engagement for making Effective LCAs

This section presents the *Typology for Design-Driven LCA Integration in LCM*, a framework based on two main factors: LCA specificity and stakeholder engagement depth. This typology categorizes LCA integration levels in innovation processes, allowing organizations to strategically adopt iterative co-design methods, align innovation objectives, and leverage LCA findings for sustainable development. Integrating design-driven participatory methods with LCAs bridges the gap between theoretical assessments and practical applications, a need identified by Clune and Lockrey (2014) and further supported by Borrion et al. (2019).

The first dimension, ***Specificity of Functional Unit and Scenario Definition***, measures how precisely the FU is defined. High specificity indicates a clear, measurable FU based on empirical data, while low specificity suggests a more generalized or broad FU, typically when the product or service scope is yet to be clearly established. The second dimension, ***Depth of Stakeholder Engagement***, reflects the extent of stakeholder involvement and the application of design methods throughout the LCA process. High engagement involves iterative co-design with stakeholders shaping decisions and scenario refinements, whereas low engagement remains largely technical, with minimal stakeholder involvement. Such engagement is crucial for making LCAs contextually relevant and actionable, addressing calls for broader stakeholder participation as suggested by Wender et al. (2014) and Sevigné-Itoiz et al. (2021).

Typology Overview and Examples

Combining these two dimensions produces four ideal-types (Dorland, 2018; Fligstein and MacAdam, 2011; King et al., 1994; Pel et al., 2020) that help guide LCA design based on organizational context. This typology supports stakeholders in selecting LCA methodologies (See Table 5) that align with organizational needs and maturity, thereby addressing a key challenge in the literature (Pryshlakivsky & Searcy, 2021).

Table 5 – typology of LCA design approaches

High engagement	**Type 2: Complex Assessment** Low specificity, high engagement. Appropriate for contexts with undefined objectives where diverse stakeholder insights increase specificity. *Example:* Municipalities seeking broad sustainability initiatives without initial clarity on impact areas	**Type 4: Design-integrated Assessment** High specificity, high engagement. This model integrates detailed data collection with intensive stakeholder engagement for highly precise, actionable outcomes. *Example:* LCA of a bespoke office chair where engineering specifications must align with user needs.

Low Engagement	**Type 1: Explorative Assessment** Low specificity, low engagement. Used for preliminary analyses to identify broad environmental hotspots without diving into specifics, ideal for early-stage sustainability strategies. *Example:* An initial LCA of a product portfolio to locate primary environmental impacts, as seen in *Design* and early *Municipality* cases.	**Type 3: Precision-Driven Assessment** High specificity, low engagement. Suitable where the product scope is clear and stakeholder input is less essential for defining FU or calibrating LCA outputs. *Example:* LCA of standardized components, like concrete tiles, where subsequent design adjustments stem from LCA results.
	Low Specificity	High Specificity

This typology guides LCA studies by aligning assessment contexts and objectives with suitable stakeholder engagement and design methods—a significant need identified in the literature. For instance, as Bisinella et al. (2021) note, defining functional units and scenarios can be challenging without stakeholder insight. Each ideal-type has unique benefits: Type 1 assessments lay foundational insights for future studies (Types 2-4). In the Municipality case, this was achieved by adding an initial diamond to the double-diamond model, allowing Type 1 to iteratively progress into Type 2 or 3. Type 4 provides high specificity with stakeholder alignment, indispensable for LCM that drives sustainable development. The Municipality case, for example, evolved through multiple ideal-types, illustrating how LCA can transition from an assessment tool to an innovation driver (Motta et al., 2018).

Selecting the Right LCA Approach

Choosing an appropriate typology is essential for effective LCA application. For example, using Type 3 methods in a low-specificity context can lead to inaccurate assumptions, as was nearly the case with *Furniture*. Studies like those by Yousefzadeh and Lloyd (2021) and Bras-Klapwijk (1999) emphasize the importance of aligning LCA assumptions with stakeholder insights to avoid premature 'closure' and ensure robustness in decision-making, underscoring the need for alignment with the typology's specificity and engagement depth.

Organizational maturity in sustainable development is critical in choosing an LCA approach. For example, organizations with limited environmental insight may benefit from a Type 1 analysis for initial hotspot identification, where engagement is less essential. Over time, organizations are encouraged to transition toward iterative, stakeholder-focused assessments (Types 2 or 3) for greater precision and alignment with sustainable innovation goals, as supported by Clune and Lockrey (2014), before going into a Type 4 assessment that ties to LCM. For LCM, i.e., the use of LCA insights to drive sustainable development, Type 4—combining high specificity and stakeholder alignment—becomes indispensable, mirroring Borrion et al. (2019) and Bras-Klapwijk's (1999) arguments for broad stakeholder engagement in impactful decision-making.

The Municipality case illustrates this progression, beginning with a Type 1 approach, advancing to Type 3, and ultimately moving toward a Type 4 iterative assessment. The

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typology not only clarifies the LCA specificity and engagement required but also maps the maturity and development timeline of an organization's sustainable practices. For LCM, i.e. the use of the developed knowledge for sustainable development, Type 4—combining high specificity and stakeholder alignment—becomes indispensable.

Implementing LCAs as Innovation Drivers

For organizations aiming to integrate LCAs into innovation, this typology provides a strategic outline, starting with a self-assessment to determine their maturity level and position within the framework. This reflective approach aligns LCA methods with organizational context, data availability, and capacity for stakeholder engagement, facilitating the most effective pathway. This approach addresses the complexity often associated with aligning LCA results with innovation strategies (Sevigné-Itoiz et al., 2021).

For companies just beginning their sustainability journey, the typology suggests that an initial Type 1 LCA approach can provide a foundational assessment. Companies with no environmental impact data, and complex enough operations to have more than one product or service, fit in this category. This preliminary LCA, often involving a basic hotspot analysis, introduces environmental considerations and helps establish a baseline understanding of impact areas, fostering greater awareness within the organization. Generic published LCAs, simplified LCA tools, and cost-based scope-3 analysis, can be used here.

For companies with an initial grasp of their environmental impact, a Type 2 or Type 3 assessment is next. Type 2 is for large complex organisations where stakeholder engagement is essential to identify areas for improvement. Type 3 is for more straightforward cases, and can serve as a more targeted, data-rich analysis that identifies specific improvement areas without extensive stakeholder input.

As the organization matures and becomes more familiar with the nuances of its environmental footprint, moving to a Type 4 LCA approach becomes viable. These approaches integrate more comprehensive, iterative engagement with stakeholders to align LCA findings with practical, actionable strategies through LCM that support long-term sustainability goals. This evolution through the typology reflects an organization's journey toward sustainable development and illustrates how LCAs evolve from compliance tools to powerful drivers of sustainable development and innovation, where iterative LCAs provide continuous insight, echoing Borrion et al. (2019) and Kjaersgaard et al. (2021) on the value of evolving LCA practices for sustainable impact.

By tailoring LCA methods to fit their maturity level and integrating insights from stakeholder engagement, organizations can bridge the gap between technical assessments and practical, impactful sustainability strategies. This typology-driven approach offers a flexible yet structured pathway for organizations to navigate their sustainability journeys, fostering a culture of continuous improvement that embeds environmental responsibility within every layer of the business

The Role of Design in the Framework

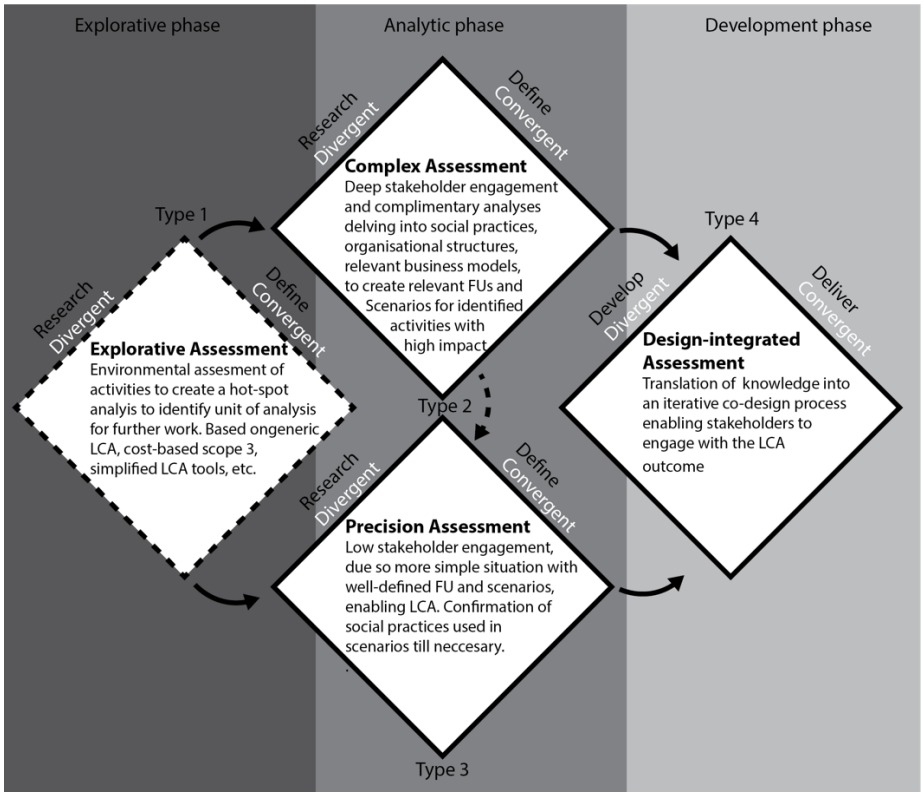


Figure 4 – The triple-diamond model. The relevance of the first diamond in the model depends on the situation. If an organisation have no environmental data available, and there a range of options that might be relevant, an explorative assessment is necessary. Depending on the outcome and the type of organisation, a process should lead to the type 2 or 3 diamond where more in-depth assessments take place. Depending on the complexity and need for detailed environmental data before progressing to type, a type to might go to a type 3 first.

Design is crucial to this framework, enabling transdisciplinary collaboration and integrating mixed-method approaches, as highlighted by Clausen et al. (2020). Iterative design processes guide the identification of knowledge needs, orchestrate relevant analyses, and translate findings into accessible, context-specific artifacts—such as models or visualizations—that facilitate meaningful stakeholder engagement. These design artifacts make complex insights comprehensible and actionable. Moreover, design provides an overarching framework for structuring the project and shaping the research design, design methods align the analytical depth of LCA with the contextual adaptability of participatory design, bridging an important gap in LCAs practical application as indicated by Clune (2014) and Motta, Issberner, and Prado (2018).

This paper was presented at ISPIM Connects Osaka – Connecting and Empowering Society, Osaka, Japan on 2-4 December 2024.
Event Proceedings: ISBN 978-952-65069-7-5

Conclusion

This paper introduces a typology for integrating LCA with LCM and PD to support sustainable business development and innovation. Our typology categorizes LCA application based on FU specificity and stakeholder engagement depth, guiding organizations in strategically aligning LCA methodologies with their sustainability maturity.

Case studies reveal that the right LCA approach—whether an initial explorative assessment or a high-specificity, iterative engagement—hinges on organizational context and goals. While initial assessments build baseline insights (Type 1), stakeholder-driven approaches (Types 3 and 4) enable actionable strategies that align LCA findings with sustainable development objectives. Importantly, design acts as a cohesive framework, facilitating transdisciplinary collaboration, stakeholder engagement, and the practical application of LCA outcomes.

Ultimately, this framework transitions LCA from a compliance-based tool to a dynamic driver of sustainable innovation, offering a structured yet adaptable pathway for organizations to embed environmental responsibility into their core strategies.

However, the article did not go into the specificity of the various complementary methods, theories, and analyses used, as it was beyond the scope of this initial article sketching out the framework. Hopefully the article did shed enough light on the considerations and process necessary to make a relevant LCA that can drive innovation in an organisation.

Acknowledgements

A special thanks to Martine Agnes Andersen; Emma Lindström Nielsen; Pauline Sofie Tøttrup; Johanne Finseth Beck; Amalie Fosvang Agesen; Thea Marie Borrell; Camilla Liborius Birkholm; Anne Albøge Olesen; Amalie Skytte Justesen; Sara Olander Lauridsen; Marie Lærke Thorball Søtofte, who have participated in the two exemplary projects, and a thanks to all the students involved in the remaining projects that have played a smaller role in the analysis.

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